

Zannatul Fardaush Tripty

📍 Dhaka, Bangladesh

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🌐 Personal Portfolio zannatul007.github.io/portfolio/

Education

Bachelor of Science (with HONORS) in Computer Science and Engineering Feb 2019 - Jun 2024
Chittagong University of Engineering and Technology (CUET) Chittagong, Bangladesh
CGPA – 3.78/4.00; Rank: 9th out of 131 students

Professional Experience

Research Assistant (Remote) Aug 2025 - Present
Elite Lab, New York, USA

- Writing research articles on explainable and interpretable AI, with a focus on large language models (LLMs).
- Supervising and mentoring junior research interns, guiding them in project design, execution, and research best practices.
- Collaborating with the core research team to advance projects in transparent, reliable, and accessible AI technologies.

Software Engineer Intern (Onsite) Feb 2025 - Apr 2025
Newroz Technologies Limited, Dhaka, Bangladesh

- Gained advanced proficiency in **Python**, specializing in data structures, file handling, error management, and object-oriented programming (OOP) to develop robust software solutions.
- Strengthened problem-solving capabilities by applying Python to complex real-world projects, incorporating OOP principles for improved code maintainability and scalability.

Industrial Trainee (Onsite) 26th Sep 2023 - 9th Oct 2023
Brain Station 23, Dhaka, Bangladesh

- Developed a solid understanding of industry practices and methodologies

Research Experience

Undergraduate Thesis

Title: Bengali Food Item Classification and Nutritional Analysis Using Deep Learning Technique
Supervisor: Dr. Muhammad Ibrahim Khan, Professor, Department of Computer Science and Engineering, CUET

- Achieved **98.85% weighted F1-score** using a **weighted ensemble of Vision Transformer models (ViT-B/16, B/32, L/16, L/32)** for Bengali food classification.
- Built and annotated a custom dataset of 6,880 Bengali food images (24 classes), expanded to 22,968 images using augmentation techniques.
- Developed a Flask-based web application for Bengali food recognition and real-time nutritional analysis from user-uploaded images.

Publications

1. **Tripty, Z.**, Nafis, M., Chowdhury, A., Hossain, J., Ahsan, S., & Hoque, M. M. (2024, March). **CUET-SentimentSillies@ DravidianLangTech EACL2024: Transformer-based Approach for Detecting and Categorizing Fake News in Malayalam Language**. In Proceedings of the Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages (pp. 245-251).[Link](#)
2. **Tripty, Z.**, Nafis, M., Chowdhury, A., Hossain, J., Ahsan, S., Das, A., & Hoque, M. M. (2024, March). **CUETSentimentSillies@ DravidianLangTech-EACL2024: Transformer-based Approach for Sentiment Analysis in Tamil and Tulu Code-Mixed Texts**. In Proceedings of the Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages (pp. 234-239).[Link](#)

Projects

1. Library Management System [|Link](#)

Stacks: Python, SQLite

- Developed a Python-based CLI system with user authentication and book/member management.
 - Implemented secure data storage using SQLite and enabled CSV import/export.
 - Applied object-oriented design principles for modular and maintainable code.
2. Food Nutrition Checker [|Link](#)

Stacks: Python, Flask, HTML

- Developed a web app for **food recognition** using an ensemble of Vision Transformers (ViT) for 24 Bangladeshi foods.
 - Calculated and scaled nutrient values and ingredient quantities from user inputs.
 - Backend with Flask and Pandas; dynamic results via HTML templates, supporting image uploads/URLs.
3. Day Care Management System [|Link](#)

Stacks: HTML, CSS, JavaScript, Bootstrap, PHP

- Developed a web-based system for daycare reservations and service bookings.
 - Implemented admin features for tracking and managing reservations.
 - Designed the user interface with HTML, CSS, JavaScript, and Bootstrap for responsiveness.
4. Cardio Checker App [|Link](#)

Stacks: Flutter, Firebase

- Developed a health app to monitor BMI and predict heart disease risks.
 - Integrated **machine learning** for risk prediction and personalized nutrition advice.
 - Used Firebase for backend services, including authentication and data storage.

Awards & Honors

Dean’s List Award for Excellent Scholarly Achievement	2019 – 2024
Chittagong University of Engineering and Technology (CUET)	Awarded for 3 consecutive academic years
University Merit Scholarship	2019 – 2024
Chittagong University of Engineering and Technology (CUET)	All 8 semesters of study
2 nd & 5 th @DravidianLangTech@EACL 2024 (Fake News Detection, Task 1 & 2)	2024
Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages @EACL 2024	
2 nd & 5 th @DravidianLangTech@EACL 2024 (Sentiment Analysis, Tamil & Tulu)	2024
Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages @EACL 2024	

Technical Skills

Programming Languages	Python, C, C++
Libraries & Packages	TensorFlow, Pandas, NumPy, Matplotlib
Frameworks	Flask, Django, React.js, Flutter
Databases	MySQL, SQLite
Tools & Platforms	Git/GitHub, VS Code, Jupyter Notebook, Android Studio, Code Blocks
Documentation	LaTeX, MS Word, MS Excel, MS PowerPoint

Standardized Test Scores

GRE: 309/340 (Quant: 162/170, Verbal: 147/170, AWA: 3.5/6)	February, 2025
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Leadership and Extracurricular Activities

Member, CUET NLP Lab	Jan 2024 - Present
Assistant Organizing Secretary(Programming), CUET Computer Club	Aug 2023 - Jun 2024
Organizing Committee Member, HULT PRIZE at CUET	Sep 2019 - Dec 2021
Executive Member, CUET Career Club	Mar 2019 - Aug 2022